

* Required

Reading 1

Read the following passage and choose the correct answer:

Designing from scratch

Reading about the latest digital enhancements in smartphones, or maybe something about advancements in green technology. Whatever that technology might be, it all started with an idea regarding engineering design.

Typically, the process of engineering design starts with a problem or a need. Therefore, it is important to go through each consideration one by one to determine the best possible solution.

(1) First, the question to be consider—whether the building process would be on an empty lot, or replacing an old building? If it's the first option, it would be obvious to fact the decision of completely tearing down the old building or utilizing the current space and renovating it for the design. The goal is to be sustainable, which means the building should be built using methods that do not completely use up or destroy natural resources. To achieve the goal, it is possible to demolish the old building in order to use new, sustainable technology; however, it is important to purchase new building material, or use the skeleton of the old building.

(2) Since the task is to build a sustainable building, it should consider sustainable energy sources. It might be easier to use non-renewable energy (which is considered less sustainable than renewable energy), because it's often more readily available. Solar panels could be an option, if the building is in a sunny area. Another option might be wind energy, generated by large wind turbines. Furthermore, it is important to think not only about generating energy, but about saving that energy. Insulation is an important factor to consider, as it helps keep the heat in during the winter and makes sure the building doesn't get too hot in the summer.

(3) Because the goal for the building is sustainability, a baseline requirement for tenants of the building should be recycling and composting. For this reason, it is important to provide recycling and composting services to the building.

(4) Imagine we are building in an area known for having a high amount of rainfall. Since the city is covered with streets and buildings, the rain isn't absorbed by the ground. This may lead to flooding. When planning our building, it is vital to try to prevent flooding by absorbing storm water runoff. Possible solutions include roof gardens, rain gardens, and flow-through planters.

(5) As previously mentioned, many decisions are involved in the design process. And these aren't even all of them! There are many more decisions involved in the construction of a new building. However, the design process is made much easier by setting specific tasks and defining constraints. This way, it would be east to focus on providing the best possible solutions.

9

"Waste Disposal" is a suitable heading for paragraph *

(4 Points)

- ☐ one
- ☐ two
- ☐ three
- ☐ four
- ☐ five

10

"Water Sources" is a suitable heading for paragraph *

(4 Points)

- ☐ one
- ☐ two
- ☐ three
- ☐ four
- ☐ five

11

"The Solution" is a suitable heading for paragraph *

(4 Points)

- ☐ one
- ☐ two
- ☐ three
- ☐ four
- ☐ five

12

"Building Materials" is a suitable heading for paragraph *

(4 Points)

- ☐ one
- ☐ two
- ☐ three
- ☐ four
- ☐ five

13

"Energy Sources" is a suitable heading for paragraph *

(4 Points)

- ☐ one
- ☐ two
- ☐ three
- ☐ four
- ☐ five

14

"Renovating" means *

(5 Points)

- ☐ of the utmost importance
- ☐ To make changes and repairs to an object so that it is back in good condition
- ☐ to be engaged in action

15

"Vital" means *

(5 Points)

- ☐ of the utmost importance
- ☐ To make changes and repairs to an object so that it is back in good condition
- ☐ to be engaged in action

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